Neuroanatomy: Brain & Spinal Cord Imaging

1. On a standard T1-weighted Magnetic Resonance Image (MRI) of the brain, what are the characteristic signal intensities of the primary tissue components?
a. CSF appears dark (hyperintense), grey matter is bright white
b. White matter appears bright (hyperintense), grey matter is intermediate (isointense), and CSF is dark (hypointense)
c. White matter is black, grey matter is bright white
d. CSF appears bright white, white matter is dark grey
e. All tissues display a completely uniform uniform intensity
2. On a T2-weighted brain MRI scan, the signal intensities invert significantly compared to T1. On a normal T2 scan, cerebrospinal fluid (CSF) inside the ventricles appears:
a. Bright white (hyperintense)
b. Jet black (hypointense)
c. Light gray
d. Dark gray
e. Translucent blue
3. When evaluating a non-contrast Computed Tomography (CT) scan of the head, acute intracranial hemorrhage (fresh blood from a stroke or trauma) characteristically presents as a localized area of:
a. Low density (hypodensity/dark)
b. High density (hyperdensity/bright white)
c. Iso-attenuation matching CSF precisely
d. Sieve-like gas shadows
e. Concentric air bubbles
4. On axial CT or MRI slices of the cerebrum, the highly important functional primary motor cortex (Brodmann area 4) can be identified structurally as the cortical ribbon lining the:
a. Precentral gyrus (anterior boundary of the central sulcus)
b. Postcentral gyrus
c. Superior temporal gyrus
d. Lingual gyrus
e. Cingulate gyrus
5. A patient presents with an inability to speak fluently but can understand spoken language perfectly (Broca's expressive aphasia). A brain MRI will most likely reveal a localized lesion in the inferior frontal gyrus of the dominant hemisphere, which corresponds to Brodmann areas:
a. 3, 1, and 2
b. 17 and 18
c. 41 and 42
d. 44 and 45
e. 9, 10, and 11
6. The primary somatosensory cortex (Brodmann areas 3, 1, and 2), which processes general somatic sensations like touch and pain from the opposite side of the body, is located on which radiological landmark?
a. Superior frontal gyrus
b. Middle temporal gyrus

c. Precentral gyrus
d. Postcentral gyrus
e. Calcarine sulcus lips

7. On a sagittal brain MRI, the primary visual cortex (Brodmann area 17) is localized deeply within the occipital lobe, mapping along the walls and lips of the:

a. Lateral sulcus
b. Central sulcus
c. Calcarine sulcus
d. Cingulate sulcus
e. Parieto-occipital fissure

8. The primary auditory cortex (Brodmann areas 41 and 42) is identified on coronal and axial brain images along the superior surface of the temporal lobe within the:

a. Angular gyrus
b. Supramarginal gyrus
c. Inferior temporal gyrus
d. Transverse temporal gyri (Heschl's gyri)
e. Uncus matrix

9. When analyzing horizontal brain slices, the white matter is organized into major systems. The massive, C-shaped bundle of commissural projection fibers that connects the right and left cerebral hemispheres is the:

a. Internal capsule
b. Fornix
c. Anterior commissure
d. Corpus callosum
e. Corona radiata

10. On axial cross-sections, the corpus callosum is divided into segments from anterior to posterior. These specific radiological components are the:

a. Head, body, and tail
b. Rostrum, genu, body (truncus), and splenium
c. Anterior horn, central body, and posterior horn
d. Alar plate and basal plate
e. Peduncle, pons, and pyramid

11. The internal capsule is a massive pathway of projection fibers visible on axial cuts. It is bounded medially by the caudate nucleus head and thalamus, and laterally by the:

a. Claustrum
b. Amygdala
c. Lentiform nucleus (globus pallidus and putamen)
d. Corpus callosum genu
e. Hippocampus body

12. On axial brain images, the internal capsule exhibits a distinct V-shape. The specific segment located at the apex of the V, between the anterior and posterior limbs, is the:

a. Rostrum
b. Genu
c. Splenium
d. Tapetum
e. Foramen of Monro boundary

13. The posterior limb of the internal capsule is a critical radiological checkpoint because it transmits high-yield descending motor tracks from the motor cortex, known as the:
a. Corticobulbar fibers
b. Spinothalamic tracts
c. Corticospinal (pyramidal) tracts
d. Dorsal columns
e. Reticulospinal paths
14. Association fibers are white matter tracts that connect different cortical areas within the same cerebral hemisphere. The major long association bundle that links the frontal, parietal, temporal, and occipital lobes is the:
a. Corpus callosum
b. Superior longitudinal fasciculus
c. Internal capsule
d. Optic radiation
e. Anterior commissure
15. Cranial nerve nuclei display precise functional components inside the brainstem. The somatic efferent (GSE) motor nucleus that supplies the extraocular lateral rectus muscle is the abducens nucleus, located radiologically within the:
a. Midbrain tectum
b. Lower pons, deep to the facial colliculus in the fourth ventricle floor
c. Upper medulla oblongata
d. Cervical spinal cord ventral horn
e. Interpeduncular fossa roof
16. The nucleus ambiguus is a high-yield column of special visceral efferent (SVE) motor neurons that supply the muscles of the pharynx and larynx via nerves IX and X. Radiologically, it is located inside the:
a. Ventral pons
b. Dorsal midbrain
c. Medulla oblongata
d. Floor of the third ventricle
e. Internal capsule posterior limb
17. General visceral efferent (GVE) parasympathetic nuclei provide autonomic secretomotor control. The superior salivatory nucleus, which regulates secretion from the submandibular and sublingual salivary glands via the facial nerve, is situated in the:
a. Medulla oblongata lower part
b. Midbrain superior colliculus level
c. Pons
d. Hypothalamus posterior box
e. Otic ganglion core
18. The Edinger-Westphal nucleus is a GVE parasympathetic center that controls pupillary constriction and lens accommodation. Radiologically, it is located within the:
a. Pons near the abducens nucleus
b. Medulla oblongata deep to the vagal triangle
c. Cerebellar vermis core
d. Midbrain, at the level of the superior colliculus deep to the oculomotor nucleus

e. Optic chiasm track

19. General somatic afferent (GSA) sensory signals from the face enter the massive sensory trigeminal nuclear complex. The segment that extends downward through the entire medulla into the upper cervical spinal cord, processing pain and temperature, is the:

a. Mesencephalic nucleus of CN V
b. Main (chief) sensory nucleus of CN V
c. Spinal nucleus of the trigeminal nerve
d. Nucleus solitarius
e. Nucleus gracilis

20. All special visceral afferent (SVA) taste fibers carried from the tongue by cranial nerves VII, IX, and X enter the brainstem to terminate within the gustatory portion of the:

a. Nucleus ambiguus
b. Nucleus solitarius (nucleus of the solitary tract)
c. Spinal nucleus of CN V
d. Main sensory nucleus of CN V
e. Edinger-Westphal nucleus

21. Cross-sectional imaging through the midbrain reveals a highly prominent, dark crescent-shaped band of dopamine-producing neurons situated between the tegmentum and the crus cerebri. This structure is the:

a. Red nucleus
b. Locus coeruleus
c. Substantia nigra
d. Periaqueductal grey
e. Inferior olive

22. On an axial brainstem MRI at the level of the upper medulla, a distinct pair of wavy, purse-like configurations of grey matter are visible laterally, responsible for cerebellar coordination. These are the:

a. Pyramids
b. Inferior olivary nuclei
c. Gracile nuclei
d. Cuneate nuclei
e. Vestibular nuclei row

23. When evaluating a T2-weighted sagittal MRI of the spine, the spinal cord is seen as a central gray column suspended inside a bright white channel containing:

a. Epidural fat cells
b. Subdural hematoma fluid
c. Cerebrospinal fluid (CSF) inside the subarachnoid space
d. Air pockets
e. Compact bone scales

24. In a normal adult subject, the spinal cord terminates apically as a tapered point known as the conus medullaris. On a sagittal lumbar spine radiograph or MRI, the conus medullaris is seen to end at which vertebral level?

a. Lower border of T10
b. Lower border of L1 or upper border of L2 vertebra
c. Lower border of L5
d. Level of the S2 sacral segment

e. Coccygeal tip

25. Descending below the conus medullaris within the lumbar cistern is a large cluster of nerve roots traveling vertically to reach their respective exit foramina, resembling a horse's tail. This radiologically prominent structure is the:

a. Filum terminale internum
b. Ligamentum nuchae
c. Cauda equina
d. Denticulate ligament
e. Ansa cervicalis pack

26. Cross-sectional images through the spinal cord reveal a distinct configuration of gray and white matter. Unlike the cerebrum, the white matter of the spinal cord is located:

a. In the absolute central core, forming a butterfly shape
b. Peripherally, surrounding the central butterfly-shaped gray matter
c. Only within the anterior horn tip
d. Exclusively inside the central canal wall
e. In patchy chaotic spots lacking column structures

27. The peripheral white matter of the spinal cord is divided radiologically into columns or funiculi. The posterior funiculus (dorsal column) is bounded between the posterior median septum and the posterior gray horn, and it contains the:

a. Lateral spinothalamic tracts
b. Lateral corticospinal tracts
c. Fasciculus gracilis and fasciculus cuneatus (dorsal columns)
d. Ventral spinocerebellar tracts
e. Anterior corticospinal tracks

28. The lateral funiculus of the spinal cord white matter contains high-yield pathways. A lesion in this specific zone on an MRI can disrupt the main descending voluntary motor tract, which is the:

a. Dorsal column pathway
b. Lateral corticospinal tract
c. Ventral spinothalamic tract
d. Rubrospinal tract
e. Vestibulospinal path

29. A 45-year-old patient presents with a loss of pain and temperature sensation in the right lower limb. A spinal cord MRI reveals a localized lesion compressing the contralateral (left) lateral white matter funiculus, disrupting the:

a. Fasciculus gracilis
b. Lateral spinothalamic tract
c. Fasciculus cuneatus
d. Posterior spinocerebellar tract
e. Corticospinal tract

30. When performing an invasive lumbar puncture to collect CSF, the clinician inserts the needle through the L3/L4 or L4/L5 interspace. Radiologically, the needle tip enters which fluid-filled space?

a. Epidural space
b. Subdural potential space
c. Lumbar cistern (subarachnoid space)
d. Central canal of the spinal cord

e. Intervertebral disc core

31. A patient presents with progressive weakness in both lower limbs, hyper-reflexia, and a positive Babinski sign. A sagittal MRI of the cervical spine reveals a herniated intervertebral disc compressing the spinal cord, a condition termed:

a. Syringomyelia
b. Cervical spondylotic myelopathy
c. Spina bifida occulta
d. Tabes dorsalis
e. Multiple sclerosis plaque

32. Syringomyelia is a classic neuroanatomical condition characterized by the formation of a fluid-filled cavity (syrinx) within the spinal cord. On a sagittal or axial MRI, the syrinx is typically seen expanding within or near the:

a. Epidural fat pad
b. Anterior longitudinal ligament
c. Central canal, destroying the anterior white commissure
d. Posterior funiculus exclusively
e. Foramen transversarium track

33. Destruction of the decussating fibers in the anterior white commissure by an expanding syrinx in syringomyelia results in a classic clinical presentation characterized by:

a. Total loss of touch and proprioception everywhere
b. Unilateral facial paralysis with hyperacusis
c. Bilateral, segmental loss of pain and temperature sensation in a "cape-like" distribution over the shoulders and upper limbs
d. Lower motor neuron paralysis of the lower limbs only
e. Total blindness in the left eye field

34. A patient undergoes an MRI of the brain following a traumatic head injury. The scan reveals a crescent-shaped (concave) collection of blood tracking along the cerebral hemisphere that does not cross the dural cranial sutures. This radiographical presentation confirms a:

a. Epidural (extradural) hematoma
b. Subdural hematoma (rupture of bridging veins)
c. Subarachnoid hemorrhage
d. Intraventricular bleeding
e. Brainstem infarction

35. An epidural (extradural) hematoma presents on a non-contrast head CT scan as a highly characteristic lens-shaped, biconvex (lentiform) hyperdense mass. This life-threatening bleeding is typically caused by a rupture of the:

a. Bridging cerebral veins
b. Middle meningeal artery (secondary to pterion trauma)
c. Internal carotid artery aneurysm
d. Great cerebral vein of Galen
e. Basilar artery trunk

36. A 52-year-old patient presents with a sudden, agonizing headache described as the "worst headache of my life," followed by neck stiffness. A non-contrast head CT reveals hyperdense blood tracking extensively within the basal cisterns, sulci, and the interhemispheric fissure. This confirms a:

a. Subdural hematoma

b. Epidural hematoma
c. Subarachnoid hemorrhage (usually from a ruptured berry aneurysm)
d. Pituitary apoplexy mass
e. Glioblastoma tumor bleed

37. When reviewing a cross-sectional MRI of the brainstem at the level of the pons, a large bundle of white matter projection fibers can be seen exiting the pons laterally to enter the cerebellum. This structure is the:

a. Superior cerebellar peduncle
b. Middle cerebellar peduncle (brachium pontis)
c. Inferior cerebellar peduncle
d. Decussation of pyramids
e. Medial lemniscus path

38. The optic radiations (geniculocalcarine tract) form a massive white matter pathway that carries visual data from the lateral geniculate body of the thalamus to the visual cortex. Fibers that loop forward and downward into the temporal lobe before heading backward are visible on images as:

a. Broca's loop
b. Wernicke's bundle
c. Meyer's loop
d. The arcuate fasciculus
e. The uncinate tract

39. A patient presents with a complete loss of vision in the upper right quadrant of both eyes (right homonymous superior quadrantanopia). This specific deficit confirms a localized lesion in the left temporal lobe disrupting:

a. The main optic tract fibers
b. The optic chiasm center
c. Meyer's loop of the optic radiations
d. The calcarine sulcus upper lip
e. The cuneus gyrus ribbon

40. Multiple sclerosis (MS) is an autoimmune demyelinating disease of the central nervous system. On a T2-weighted or FLAIR MRI of the brain, MS lesions present characteristically as bright white patches (plaques) perpendicular to the lateral ventricles, known as:

a. Calcar avis lips
b. Dawson's fingers
c. Herring bodies
d. Perikymata rings
e. Brain sand tracks

41. The prefrontal cortex (Brodmann areas 9, 10, 11, and 12) occupies the anterior-most portion of the frontal lobe. A tumor infiltrating this region on an MRI will manifest clinically as a profound alteration in:

a. Primary visual acuity
b. Auditory frequency discrimination
c. Personality, social behavior, judgment, and executive function
d. Horizontal eye scanning movements
e. Fine motor control of the toes

42. Wernicke's area (Brodmann area 22) is a critical functional zone responsible for the comprehension of spoken language. On a brain MRI, it is localized in the posterior part of the:
a. Superior frontal gyrus
b. Superior temporal gyrus (dominant hemisphere)
c. Inferior parietal lobule
d. Lingual gyrus
e. Cingulate cortex ribbon
43. A patient presents with fluent but completely meaningless speech, accompanied by a total inability to understand spoken commands (Wernicke's receptive aphasia). The lesion maps to Brodmann area:
a. 4
b. 44
c. 22
d. 17
e. 41
44. The association tract that directly connects Wernicke's area to Broca's area, allowing for coordinated language expression and repetition, is a bundle of white matter called the:
a. Corpus callosum tapetum
b. Anterior commissure
c. Arcuate fasciculus
d. Internal capsule genu
e. Stria terminalis channel
45. On coronal brain sections, the thin, vertical sheet of tissue that stretches between the corpus callosum and the fornix, separating the anterior horns of the right and left lateral ventricles, is the:
a. Lamina terminalis
b. Septum pellucidum
c. Tentorium cerebelli
d. Falx cerebri
e. Medial longitudinal fissure
46. When evaluating an axial CT scan of the brain, the pineal gland is frequently seen as a small, bright white hyperdense dot in the midline, posterior to the third ventricle. This hyperdensity is normal and represents:
a. An acute localized hemorrhage
b. A highly malignant pinealoblastoma tumor
c. Physiological calcification of the pineal gland (brain sand)
d. An artifact caused by the skull base bone
e. A dense collection of cerebrospinal fluid
47. A patient experiences an ischemic stroke involving the left internal capsule posterior limb. Cross-sectional imaging will demonstrate tissue damage in this zone, and the patient will present clinically with:
a. Total loss of vision in both eyes
b. Pure motor hemiplegia affecting the right (contralateral) side of the face and body
c. Expressive aphasia with normal limb movements
d. Loss of taste on the posterior tongue
e. Transient facial paralysis sparing the forehead wrinkles

48. On high-resolution axial MRIs of the midbrain, a small, circular fluid-filled channel is visible exactly in the midline, surrounded by the periaqueductal grey matter. This structural channel is the:

a. Foramen of Monro
b. Fourth ventricle
c. Cerebral aqueduct (of Sylvius)
d. Foramen of Magendie
e. Central canal row

49. The anterior corticospinal tract is a small bundle of descending motor fibers located within which funiculus of the spinal cord white matter?

a. Posterior funiculus
b. Lateral funiculus
c. Anterior funiculus
d. Dorsal columns
e. Lissauer's tract

50. An MRI of the brain reveals a benign tumor (meningioma) compressing the olfactory groove on the cribriform plate. This mass will result in a progressive, unilateral loss of which special sense?

a. Vision (amaurosis)
b. Hearing (anacusis)
c. Smell (anosmia)
d. Taste (ageusia)
e. Balance (vertigo)